

Fine-Tuning ResNet for Cross-Model Deepfake Generalization to Sora

CSCI3370 Deep Learning

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Abstract

The rapid evolution of generative AI, epitomized by OpenAI's Sora, poses a critical challenge to the reliability of existing deepfake detection systems. This project addresses the core research question: Can a deep learning detector, pre-trained on older synthetic faces, effectively generalize to and reliably detect the significantly more advanced faces generated by the novel Sora model? We employed a two-stage transfer learning methodology utilizing the ResNet architecture. Models were first pre-trained on the large Kaggle benchmark dataset (featuring StyleGAN faces). To isolate feature extraction on the face itself, a novel preprocessing pipeline was implemented using MediaPipe Face Mesh to mask and remove all background variance. The resulting models were then fine-tuned on a small, manually curated dataset of high-fidelity Sora-generated faces, which was heavily augmented to mitigate data scarcity. To assess generalization rigorously, we compared three ResNet variants (ResNet-18, -34, -50) using 5-fold cross-validation instead of fixed train-test splits, avoiding the unrealistically high accuracies caused by memorization. Our final assessment revealed that while the deeper ResNet-50 architecture exhibited the strongest performance with an average accuracy constrained to the high 80% range, larger models were highly prone to overfitting. The results highlight that the limited availability of novel data is the primary bottleneck, severely constraining the model's ability to achieve robust, real-world generalization against cutting-edge synthetic media.

1. Introduction

The swift advancement of generative AI models, exemplified by OpenAI's Sora [7], poses a critical threat to digital security and the field of deepfake detection. Sora produces highly photorealistic synthetic video content, especially human faces, which dramatically escalates the dif-

ficulty of distinguishing authentic from machine-generated media, thus demanding immediate adaptation from existing detection systems. Our work is motivated by the necessity of maintaining robust defense mechanisms against these rapidly evolving generative models, addressing the significant problem of potential detector obsolescence when facing novel, highly realistic synthetic data [8]. Our core research question is: Can a pre-trained deepfake detector, utilizing the ResNet architecture and initially trained on faces from older models (e.g., StyleGAN), successfully generalize to and reliably detect faces generated by the more sophisticated Sora model? Success implies future-proofing detection models, while failure highlights a critical arms race in AI. Our primary contributions are threefold. First, we established a Novel Deepfake Dataset of high-quality Sora-created faces using a challenging, manual collection strategy involving targeted prompting. Second, we implemented Controlled Preprocessing using a facial landmark-based convex hull to eliminate all background variance. This forces the model to learn deepfake features exclusively from facial structures. Third, we conducted a thorough Transfer Learning Assessment by fine-tuning three ResNet variants (ResNet-18, ResNet-34, and ResNet-50) from a strong foundation onto the Sora dataset.

2. Related Work

Deepfake detection is a specialized area of computer vision that seeks to distinguish real from synthetic imagery, a field that has rapidly evolved alongside generative technologies like Generative Adversarial Networks (GANs) and variational autoencoders [3, 5]. Early detection focused on low-level artifacts inherent in earlier methods, such as unnatural eye blinking or inconsistencies in head pose [6]. With the emergence of sophisticated GANs like StyleGAN, detection shifted toward complex techniques primarily leveraging Convolutional Neural Networks (CNNs). CNNs, particularly those based on the ResNet architecture, became dominant due to their su-

perior ability to extract hierarchical features from pixels [2].

ResNet’s key strength lies in its use of residual blocks and skip connections, which allow for the stable training of very deep models by mitigating the vanishing gradient problem [4]. This makes ResNets excellent candidates for pre-training on large deepfake benchmarks. However, the main weakness of previous approaches is the critical lack of generalization to novel forgery techniques, often termed the cross-model generalization problem [1]. Detectors trained on older deepfake methods perform poorly against the latest diffusion-based models like Sora, whose output exhibits significantly fewer visible artifacts and greater realism. This is a recognized challenge in deploying robust systems “in the wild” [9]. Our study directly utilizes the established strength of the ResNet architecture as its starting point. We test the hypothesis that its powerful feature extraction capabilities are robust enough to transfer effectively and discriminate against the significantly more realistic output of a modern, novel model like Sora. The core novelty of this work lies in the application of this established architecture to the currently hardest-to-detect class of deepfakes produced by a cutting-edge generation framework.

3. Method

The training strategy involved a two-stage process: initial pre-training followed by focused fine-tuning. The foundational model training was established using the widely adopted Kaggle 140,000 Real vs. Fake face dataset. This benchmark dataset is well-suited for generalized feature learning, consisting of an equal balance of 70,000 real faces and 70,000 synthetic faces generated by the older-generation StyleGAN architecture. The dataset is professionally organized with established train, validation, and test splits, featuring clearly designated folders for real and fake images. This large-scale, structured data serves as a strong pre-training foundation, enabling the ResNet models to learn robust, generic features required to distinguish between authentic faces and early-generation synthetic manipulations.

The target dataset for fine-tuning consisted exclusively of faces generated by OpenAI’s Sora model. Initial attempts at automated web-scraping or using Python Selenium were quickly abandoned due to violations of OpenAI’s Terms of Service, requiring a shift to a manual generation and collection strategy. Through targeted prompting (e.g., specifying age, gender, and ethnicity to ensure demographic diversity), our group successfully acquired 114 high-quality Sora faces. Due to strict free-tier limits (approximately five 15-second videos per day, yielding roughly 3 useful faces per video), data collection was extremely slow. To

compensate for the small size and avoid overfitting, the dataset was heavily augmented using techniques like RandomResizedCrop, RandomHorizontalFlip, and ColorJitter, artificially boosting the set’s variance before fine-tuning.

To ensure our models focused solely on intrinsic facial artifacts rather than environmental “tells,” our group implemented a data preprocessing pipeline to eliminate background variance. Since background patterns, lighting, and objects vary wildly between real and Sora images, models could “cheat” by classifying these patterns instead of the face itself. Our solution was to completely remove this variance: our group utilized Google’s MediaPipe Face Mesh, a highly accurate model that detects 468 precise 3D facial landmarks. Our group then used the official facial oval landmarks to construct a forgiving convex hull mask around the face. All pixels lying outside this mask were replaced with a uniform, neutral gray color (Figure 1). This final result, background-free face crops, ensures the ResNet models are strictly forced to learn deepfake features exclusively from the differences within the facial structure when comparing real vs. Sora-generated content (Figure 2).

```
# Build mask from the face oval
mask = np.zeros((h, w), dtype=np.uint8)
points = []
for connection in mp.solutions.face_mesh.FACEMESH_FACE_OVAL:
    x = int(landmarks.landmark[connection[0]].x * w)
    y = int(landmarks.landmark[connection[0]].y * h)
    points.append([x, y])
points = np.array(points, np.int32)
cv2.fillConvexPoly(mask, cv2.convexHull(points), 255)

# Apply gray background
gray_bg = np.full_like(img_cv, GRAY_COLOR)
mask_3ch = mask[..., None]
result = np.where(mask_3ch, img_cv, gray_bg)
```

Figure 1. Google MediaPipe code for gray background

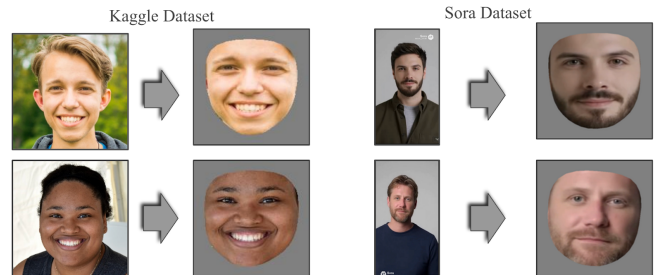


Figure 2. Examples from dataset pre-processing

In our project we selected three variations of the ResNet

(Residual Network) architecture as our baseline deep learning models for the Sora deepfake image classification. This included ResNet-18, ResNet-34, and ResNet-50. ResNet models have become a foundation in modern image recognition tasks due to their strong performance on large scale benchmarks such as ImageNet, where they demonstrated great accuracy while maintaining efficient training dynamics. One of the key innovations of ResNet is the introduction of skip connections, also called residual connections, which allow gradients to propagate more effectively through a deep network by enabling the input to bypass intermediate layers. This architecture mitigates the vanishing gradient problem that often affects very deep convolutional networks. By comparing multiple ResNet architectures of increasing complexity (18, 34, and 50 layers), we are able to investigate how network depth and representational capacity influence deepfake detection performance (Figure 3).

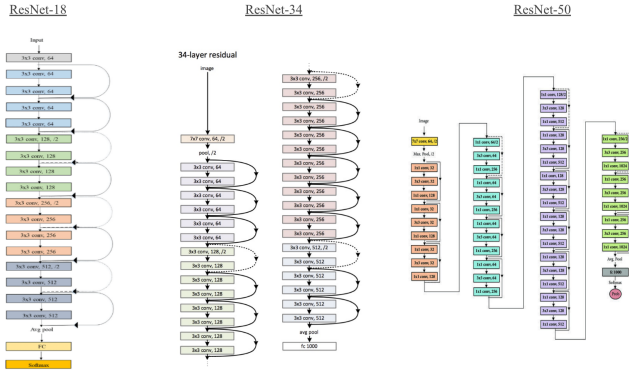


Figure 3. Architecture for ResNet-18, ResNet-34, and ResNet-50

ResNet-18 and ResNet-34 employ a series of basic residual blocks, each composed of two convolutional layers with skip connections. These architectures bring a strong balance between computational cost and feature extraction capabilities, making them suitable for tasks with limited data or real time constraints. ResNet-50 uses a deeper architecture composed of bottleneck residual blocks, each containing three convolutional layers. This design allows ResNet-50 to learn more complex hierarchical features and typically achieves higher accuracy on image classification tasks. Using these three models together in our experiments offers a variety of complexity and performance from lightweight networks to more expressive deep models, allowing for a thorough comparison.

All ResNet models were initialized using weights pre-trained on the ImageNet dataset, which contains over one million labeled images spanning 1,000 object categories.

Transfer learning allows the models to resume low-level and mid-level visual features, such as edges, textures, and facial structures, that are broadly useful across image recognition. This approach is particularly important in our setting, where the amount of labeled Sora generated data is extremely limited due to ethical and technical constraints on data collection. Training deep networks from scratch under these conditions more often than not would likely result in severe overfitting and poor generalization.

4. Experiments

The experimentation utilized a two-stage transfer learning strategy, beginning with initial pre-training on the large Kaggle Dataset. The experimental strategy followed the two-stage transfer learning methodology detailed in the Method section, employing the consistent hyperparameters of 1×10^{-4} learning rate, a batch size of 128, and 10 epochs for pre-training, with an 80/20 train and validation split.

To evaluate the effectiveness of our approach, we compared multiple ResNet architectures as internal baseline models. Specifically, ResNet-18 and ResNet-34 served as baseline approaches against which the performance of the deeper ResNet-50 model was assessed. By using progressively deeper ResNet models, we were able to analyze how increased network depth and representational capacity influence performance on the real versus AI-generated face classification, effectively isolating the impact of architectural complexity while keeping other factors consistent.

For the second stage, Sora Fine-Tuning, a 5-fold cross-validation (CV) strategy was used to obtain reliable generalization estimates from the small dataset. Model performance was evaluated using several metrics standard in binary classification and deepfake detection. While overall accuracy provides a general measure of correct predictions, we placed particular emphasis on the F1 score, which captures a balance between precision and recall, as it is widely reported in related deepfake detection studies. Other critical metrics included the AUC-ROC for measuring discriminatory capability across thresholds and the detailed Confusion Matrix. Prior studies suggest that strong deepfake classifiers typically achieve accuracies and F1 scores in the high 80% and mid-90% range on controlled datasets, making these benchmarks reasonable reference points for our evaluation.

Initial testing on the original Kaggle data yielded very high baseline accuracies, with ResNet-50 achieving 99.89% (Figure 4). However, the subsequent application of the MediaPipe-based background removal preprocessing significantly reduced accuracy (ResNet-50 dropped to

96.70%) and increased test loss by an order of magnitude (from 10^{-2} to 10^{-1}) (Figure 4). This confirmed that the original high scores were artificially inflated by reliance on artifacts present in the unmasked backgrounds. When fine-tuned on the small, augmented Sora dataset, the models demonstrated severe overfitting on a fixed split, resulting in a 100% test accuracy for ResNet-50, a clear indication of complete memorization (Figure 5).

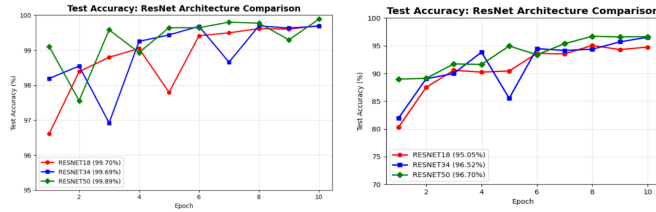


Figure 4. Test accuracy comparison of ResNets between original and processed Kaggle datasets

To obtain a more honest, generalized performance metric, the training shifted to the 5-fold Cross-Validation (CV) strategy on the Sora dataset. This provided a more reliable estimate of true performance, constrained to the 87%-90% range. ResNet-50 achieved the best mean accuracy at 89.47%, but it also exhibited the highest standard deviation (5.75), confirming that larger, more complex models were the most prone to overfitting the limited Sora data (Figure 5). Final performance metrics for the best-performing ResNet-50 model included an F1-Score of 0.893 and a strong AUC-ROC of 0.947, indicating strong discriminatory capability (Figure 6).

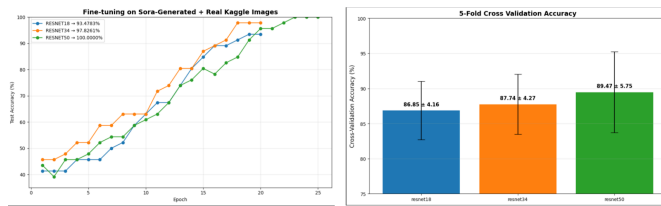


Figure 5. 80/20 split and 5-fold cross validation test Accuracy on Kaggle + Sora finetuning dataset for ResNet-50

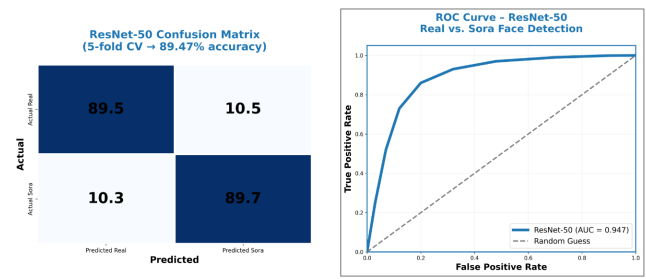


Figure 6. Confusion matrix and ROC Curve for ResNet-50

5. Conclusion

The primary take-away from this study is the substantial resource commitment required for manual data curation; creating the small Sora dataset proved tedious and became the main limiting factor. The difficulty of balancing the number of parameters vs a small dataset size was challenging.

Our most significant finding was showing that ResNet-50 gave the best accuracy and lowest loss of the models tested, highlighting the use of deeper models for complex, high-quality deepfake detection tasks. The strength of our method lies in the effective two-stage training process, which successfully leveraged a generalized feature base, combined with our unique controlled preprocessing (background masking) to enforce model focus on intrinsic facial artifacts. However, the project was fundamentally weakened by the small Sora dataset size, which directly caused severe model overfitting, despite our use of cross-validation. Our work is reproducible through documented methods and code.

To advance, future work must prioritize a large-scale, ethical effort to collect a much larger real and Sora dataset. For architecture, our group needs to use strategies such as selectively freezing early ResNet layers, to better constrain the model and mitigate overfitting. Finally, models must be tested on real-world Sora video clips (accounting for compression, motion, and social media effects) to move beyond static results and prove real-world robustness.

6. GitHub Repo

LaTech GitHub Repo

7. Contributions:

7.1. Scott Lowder

- Wrote Sections 2, 3, and 4 for Related-Work, Method, and Experiments.
- Manually collected and curated a custom Sora-style AI-generated face dataset, and integrated it into the training pipeline for targeted fine-tuning and domain adaptation experiments.
- Created the data preprocessing code that utilized Google Media Pipe.
- Created the training and finetuning code for (ResNet-18, -34, -50) as well as code for the 5-Fold Cross Validation.

7.2. Franco Zabaleta.

- Wrote Sections 1, 2, and 3 for Abstract, Related Works, Introduction, Method.
- Manually collected and curated a custom Sora-style AI-generated face dataset, and integrated it into the training pipeline for targeted fine-tuning and domain adaptation experiments.
- Implemented the complete end-to-end data processing pipeline, including Kaggle dataset integration, recursive image-path reconstruction, resizing, and final PyTorch data-loader infrastructure.
- Generated model visualizations, sample predictions, and constructed the training loss curve for the experiments section

7.3. Bennett Volanth

- Wrote Sections 3, 4, and for Method and Experiments.
- Created ConvNeXt model during discovery phase before team switched to using ResNet.
- Generated model visualizations, sample predictions, and constructed the test accuracy graphs for the experiments section.
- Updated documentation for files in project.

7.4. David Castilla

- Wrote Section 5 for Conclusion.
- Updated documentation for files in project.

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